Siddhesh Pawar

Melbourne, VIC | Temporary Graduate Visa (Subclass 485), valid until 11 August 2029 | LinkedIn | siddhesh.pawar.1397@gmail.com

PROFESSIONAL SUMMARY

Data Science and Engineering professional with industry experience in building data pipelines, NLP/GenAI solutions, and ML models in financial services and consulting. Recognised with a CTO Spot Award for automating risk-reporting workflows that cut turnaround from one week to fifteen minutes. Currently leading the data science and engineering workstream on StrideSafe, an end-to-end heat-safety platform powering a Cool Routing engine over 76,416 street segments.

EDUCATION

Monash University — Master of Data Science, Melbourne **Jul 2024 – Jun 2026**
GPA 3.438/4.00 | WAM 80.125 | Coursework: Data Processing for Big Data, Data Analysis for Semi-Structured Data, Data Wrangling, Databases, Data Exploration and Visualisation

University of Mumbai — B.E. Electronics & Telecommunication **Aug 2018 – Jun 2022**
CGPA 8.97/10.00

TECHNICAL SKILLS

Languages: Python, R, SQL
ML / AI: PyTorch, scikit-learn, Natural Language Processing, Recommendation Systems
Data Engineering: PySpark, Apache Kafka, Apache Airflow, ETL pipeline design, data quality controls
Databases: MS SQL Server, MySQL, Oracle
Visualisation: Power BI, Tableau, Apache Superset, R Shiny
Cloud & Tools: AWS, Git, Jupyter Notebook, VS Code, SQL Server Management Studio, Miro Board
Design Thinking: User Persona, Empathy Map, Lotus Blossom, User Journey Map

PROFESSIONAL EXPERIENCE

Data Analyst — NeoGrowth Credit Pvt. Ltd., Mumbai **Apr 2023 – May 2024**
Engineering team — supported Risk, Customer Support, Finance, and Technology departments

- **Automated regulatory credit risk reporting** (Historical Static Pool, Expected Credit Loss) for the Risk Department using Python and MS SQL Server, cutting turnaround from **1 week to 15 minutes** — recognised with a **CTO Spot Award** for improving traceability for internal audit and compliance reviews.
- **Built NLP and GenAI solution** for inbound email content extraction and classification; integrated OpenAI GPT models to auto-generate conversational reply bodies, serving as organisational point of contact for OpenAI services.
- **Migrated 3 business-critical reports** from a legacy reporting application to Apache Superset and **monitored migration of 20+ reports** end-to-end; developed SQL-based data quality controls to detect anomalous financial records across the loan book.
- **Designed executive compliance dashboards** for CEO, CTO, and Risk Vice Presidents tracking audit-finding remediation across severity bands; built a **machine learning credit risk model** supporting risk-based customer evaluation and data-driven lending decisions.

Associate Software Engineer — Go Digital Technology Consulting, Mumbai **Jan 2023 – Mar 2023**
Client: Duke Energy (Fortune 500 US energy utility)

- Delivered on an **energy consumption forecasting** engagement: orchestrated data migration from on-premises systems to AWS and developed a SQL-based data transfer framework to standardise hand-offs between engineering teams.
- Implemented functional business logic changes in production **PySpark** scripts (weather related data) supporting the forecasting workflow.

ACADEMIC PROJECT

Data Science & Engineering Lead — StrideSafe *(Monash Industry Experience Project)* **Feb 2026 – Present**
Cross-functional capstone team across data science, IT, and BIS | IP owned by the team

- Led end-to-end data architecture for a heat-safety platform covering the City of Melbourne LGA, integrating urban forest canopy, building footprints, OpenStreetMap, Open-Meteo weather, and microclimate sensor data into a deterministic per-edge **thermal scoring model** across **76,416 street segments**.
- Designed and implemented a **Cool Routing engine** combining **54,000+ tree canopy records** with physics-based building **shadow modelling** (pvlib sun-position, GeometryCollection optimisation) to recommend the coolest walking and cycling route between any two points in the LGA.
- Engineered a monthly precomputation pipeline for static spatial geometry, enabling sub-second route recommendations at query time on commodity infrastructure.
- Built the **heat-risk recommendation engine** combining simplified Wet Bulb Globe Temperature (sWBGT) with activity-intensity (MET) values.

Global Trade & Renewable Energy Adoption Dashboard — R Shiny Application

- Built a multi-section dashboard analysing **~$65.9 Trillion USD** of global trade across 25 reporter countries (2000–2024) from UN Comtrade and OECD;
- Implemented a choropleth trade-intensity map, top-100 export-flow network with year-over-year animation, k-means clustering of countries by renewable-vs-fossil growth composition (tunable k and trailing window), and a Sankey diagram for bilateral trade-profile comparison.

EXTRACURRICULAR

- **Second Prize, TCS Design Thinking Competition** (Tata Consultancy Services, 2021) — Designed a Clean City Challenge solution using user persona, empathy map, and user journey mapping.
- **Volunteer, Monash Student Association** (Sept 2024 – Present) — Coordinated check-in and logistics for student events of 200+ attendees.